

Exercise 2

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Problem 2.1a.

- Solution.*
1. $\mathbb{P}_x(X_1 = y, X_0 = x) = P(X_1 = y, X_0 = x \mid X_0 = x) = P(X_1 = y \mid X_0 = x) = p_{x,y}$
 2. $\mathbb{P}_x(X_1 = y) = P(X_1 = y \mid X_0 = x) = p_{x,y}$
 3. $\mathbb{P}_x(X_1 = x, X_2 = y) = P(X_1 = x, X_2 = y \mid X_0 = x) = p_{x,x}p_{x,y} \neq p_{x,y}$
 4. $\mathbb{P}_x(X_1 = x, X_0 = y) = P(X_1 = x, X_0 = y \mid X_0 = x) = 0$
 5. $\mathbb{P}_{\delta^x}(X_1 = y) = \delta^x(x)p_{x,y} = p_{x,y}$

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Problem 2.1b.

- Solution.*
1. $\mathbb{P}_x(X_{n+2} = z, X_{n+1} = y \mid X_n = x) = p_{x,y}p_{y,z} \neq \mathbb{P}_x(X_{n+2} = z, X_{n+1} = y, X_n = x)$
 2. $\mathbb{P}_x(X_{2n+2} = z, X_{2n+1} = y, X_{2n} = x \mid X_n = x) = \mathbb{P}_x(X_{n+2} = z, X_{n+1} = y, X_n = x)$ by Homogeneity.
 3. $p_{x,x}^{(n)}p_{x,y}p_{y,z} = \mathbb{P}_x(X_n = x, X_{n+1} = y, X_{n+2} = z)$
 4. $p_{x,y}p_{y,z} \neq p_{x,x}^{(n)}p_{x,y}p_{y,z} = \mathbb{P}_x(X_n = x, X_{n+1} = y, X_{n+2} = z)$
 5. $\sum_{u_1, \dots, u_{n-1} \in S} p_{x,u_1} \cdots p_{u_{n-1},x}p_{x,y}p_{y,z} = p_{x,x}^{(n)}p_{x,y}p_{y,z} = \mathbb{P}_x(X_n = x, X_{n+1} = y, X_{n+2} = z)$

■

Problem 2.1c.

- Proof.*
1. $\mathbb{P}_1(X_2 = 3) = \frac{1}{4} \neq 0$
 2. $\mathbb{P}_1(X_3 = 3)$: 3 is 2 away from 1. So the number of +1 or r has to be 2 greater than the number of -1 or l . So $2l + 2 = r + l = 3 \implies l = \frac{1}{2}$. But l has to be a natural number. Thus this is not possible so $\mathbb{P}_1(X_3 = 3) = 0$
 3. $\mathbb{P}_1(X_4 = 3) > \mathbb{P}_1(X_4 = 3, X_3 = 2, X_2 = 1, X_1 = 0) \neq 0$
 4. $\mathbb{P}_1(X_5 X_6 < 0)$: This means $X_5 > 0$ and $X_6 < 0$ or $X_6 > 0$ and $X_5 < 0$. Thus $|X_5 - X_6| \geq 2$ and thus given the transition probability is 0. Thus $\mathbb{P}_1(X_5 X_6 < 0) = 0$.
 5. $\mathbb{P}_1(X_{2n+1} = 0)$ for $n \in \mathbb{N}$: $\mathbb{P}_1(X_{2n+1} = 0) > \mathbb{P}_1(X_1 = 0, X_2 = 1, X_3 = 0, \dots, X_{2n} = 1, X_{2n+1} = 0) = (\frac{1}{2})^{2n+1} > 0$

□

Problem 2.2.

Proof. The transition matrix is P .

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}$$

The characteristic polynomial is $\chi_P(\lambda) = -(\lambda^3 - \lambda^2 + \frac{1}{4}\lambda - \frac{1}{4})$. 1 is always an eigenvalue of stochastic matrices. thus $\chi_P(\lambda) = -(\lambda - 1)(\lambda^2 + \frac{1}{4}) = -(\lambda - 1)(\lambda - \frac{1}{2}i)(\lambda + \frac{1}{2}i)$.

For some $\alpha, \beta, \gamma \in \mathbb{C}$, $p_{a,a}^{(n)} = \alpha + \beta(\frac{i}{2})^n + \gamma(\frac{-i}{2})^n$.

$$p_{a,a}^{(0)} = \alpha + \beta + \gamma = 1$$

$$p_{a,a}^{(1)} = \alpha + \frac{i}{2}(\beta - \gamma) = 0$$

$$p_{a,a}^{(2)} = \alpha - \frac{1}{4}(\beta + \gamma) = 0$$

After some algebraic manipulations you see $\alpha = \frac{1}{5}$, $\beta = \frac{2}{5} + \frac{1}{5}i$, and $\gamma = (\frac{2}{5} - \frac{1}{5}i)$. Now we simplify.

$$\begin{aligned} p_{aa}^{(n)} &= \frac{1}{5} + \left(\frac{2}{5} + \frac{1}{5}i\right) \left(\frac{i}{2}\right)^n + \left(\frac{2}{5} - \frac{1}{5}i\right) \left(\frac{-i}{2}\right)^n \\ &= \frac{1}{5} + \frac{1}{2^n} \left(\frac{2}{5} (i^n + (-i)^n) + \frac{1}{5} (i^{n+1} + (-i)^{n+1})\right) \\ &= \frac{1}{5} + \frac{1}{2^n} \left(\frac{2}{5} (e^{i\frac{\pi}{2}n} + e^{-i\frac{\pi}{2}n}) + \frac{1}{5} (e^{i\frac{\pi}{2}(n+1)} + e^{-i\frac{\pi}{2}(n+1)})\right) \\ &= \frac{1}{5} + \frac{1}{2^n} \left(\frac{4}{5} \cos\left(\frac{\pi n}{2}\right) + \frac{2}{5} \cos\left(\frac{n\pi}{2} + \frac{\pi}{2}\right)\right) \\ &= \frac{1}{5} + \frac{1}{2^n} \left(\frac{4}{5} \cos\left(\frac{\pi n}{2}\right) + \frac{2}{5} \sin\left(\frac{n\pi}{2}\right)\right) \end{aligned}$$

□

Problem 2.3.

Proof. Let μ be a distribution. $\forall A \subseteq S$, let $\mu(A) = \mathbb{P}(X_0 \in A)$. Define $P : S \times S \rightarrow [0, 1]$ where $\forall x, y \in S$, $P(x, y) = \mathbb{P}(X_1 = y \mid X_0 = x)$.

$$\begin{aligned}
\mathbb{P}(X_n = x_n, \dots, X_0 = x_0) &= \mathbb{P}(X_0 = x_0) * \mathbb{P}(X_n = x_n, \dots, X_1 = x_1 | X_0 = x_0) \\
&= \mathbb{P}(X_0 = x_0) * \prod_{k=1}^n \mathbb{P}(X_k = x_k | X_0 = x_0, \dots, X_{k-1} = x_{k-1}) \\
&= \mathbb{P}(X_0 = x_0) * \prod_{k=1}^n \mathbb{P}(X_k = x_k | X_{k-1} = x_{k-1}) \\
&= \mu(x_0) * \prod_{k=1}^n P(x_{k-1}, x_k)
\end{aligned}$$

Thus $X \sim MC(\mu, P)$. □

Problem 2.4.

Proof. Same distribution:

$$\begin{aligned}
\mathbb{P}_0(Z' = k) &= \mathbb{P}_0\left(\sum_{n=10}^{20} 1_{X_n=X_{10}} = k\right) \\
&= \mathbb{P}_0\left(\sum_{n=0}^{10} 1_{X_{n+10}=X_{10}} = k\right) \\
&= \sum_{x \in S} \mathbb{P}_0(X_{10} = x) \mathbb{P}_0\left(\sum_{n=0}^{10} 1_{X_{n+10}=x} = k \mid X_{10} = x\right) \\
&= \sum_{x \in S} \mathbb{P}_0(X_{10} = x) \mathbb{P}\left(\sum_{n=0}^{10} 1_{X_{n+10}=x} = k \mid X_{10} = x, X_0 = 0\right) \\
&= \sum_{x \in S} \mathbb{P}_0(X_{10} = x) \mathbb{P}\left(\sum_{n=0}^{10} 1_{X_{n+10}=x} = k \mid X_{10} = x\right) \\
&= \sum_{x \in S} \mathbb{P}_0(X_{10} = x) \mathbb{P}\left(\sum_{n=0}^{10} 1_{X_n=x} = k \mid X_0 = x\right) \\
&= \sum_{x \in S} \mathbb{P}_0(X_{10} = x) \mathbb{P}_x(Z = k) \\
&= \sum_{x \in S} \mathbb{P}_0(X_{10} = x) \mathbb{P}_0(Z = k) \\
&= \mathbb{P}_0(Z = k) \left(\sum_{x \in S} \mathbb{P}_0(X_{10} = x)\right) = \mathbb{P}_0(Z = k)
\end{aligned}$$

□

Problem 2.5a.

Proof. For $x \in [-N, N] \cap \mathbb{Z}$,

$$\begin{aligned}\mathbb{P}_x(H_{-N,N} \leq N) &\geq \mathbb{P}_x(H_{-N,N} \leq \min(|N-x|, |-N-x|)) \\ &\geq \mathbb{P}_0(H_{-N,N} \leq N) \\ &\geq 2^{-N}\end{aligned}$$

Thus $\mathbb{P}_x(H_{-N,N} > N) \leq 1 - 2^{-N}$. Also note $\mathbb{P}_x(H_{-N,N} > N) \leq \mathbb{P}_0(H_{-N,N} > N)$.

Inductive Hypothesis: $\mathbb{P}_0(H_{-N,N} > kN) \leq (1 - 2^{-N})^k$ for some $k \in \mathbb{N}$.

Inductive Step:

$$\begin{aligned}\mathbb{P}_0(H_{-N,N} > (k+1)N) &\leq (1 - 2^{-N})^k \sum_{x \in [-N, N]} \mathbb{P}_x(H_{-N,N} > N) \mathbb{P}_0(X_{kN} = x) \\ &\leq \mathbb{P}_0(H_{-N,N} > N) (1 - 2^{-N})^k \\ &\leq (1 - 2^{-N})^{k+1}\end{aligned}$$

Thus by induction $\mathbb{P}_0(H_{-N,N} > kN) \leq (1 - 2^{-N})^k$.

Expectation:

$$\begin{aligned}E_0[H_{-N,N}] &= \sum_{n=1}^{\infty} \mathbb{P}_0(H_{-N,N} > n) \\ &\leq \sum_{k=0}^{\infty} N \mathbb{P}_0(H_{-N,N} > kN) \\ &\leq N \sum_{k=0}^{\infty} (1 - 2^{-N})^k \\ &\leq \frac{N}{1 - (1 - 2^{-N})} = N \cdot 2^N\end{aligned}$$

□

Problem 2.5b.

Proof. $\forall x \in [-N, N] \cap \mathbb{Z}$, let $m = \min(|N-x|, |-N-x|)$

$$\begin{aligned}E_x(H_{-N,N}) &\leq P(H_{-N,N} \leq m)m + \sum_{y \in [-N, N] \cap \mathbb{Z}} \mathbb{P}_x(X_m = y)(E_y(H_{-N,N}) + m) \\ &\leq m + E_0(H_{-N,N}) + m \\ &\leq 2m + N \cdot 2^N \\ &\leq 2N + N \cdot 2^N \\ &\leq 2N(1 + 2^{N-1}) < \infty\end{aligned}$$

□

Problem 2.5c.

Proof. Let $f : [-N, N] \cap \mathbb{Z} \rightarrow [0, \infty)$ where $x \rightarrow E_x[H_{-N, N}]$. Note for $x \in (-N, N) \cap \mathbb{Z}$, $f(x) = 1 + \frac{1}{2}f(x-1) + \frac{1}{2}f(x+1)$

Let $y = x + 1$.

Thus

$$f(x-1) - 2f(x) + f(x+1) = f(y) + 2f(y-1) + f(y-2) = -2$$

Homogeneous Part: The characteristic equation is

$$x^2 - 2x + 1 = 0$$

The root is $x = 1$ with multiplicity 2.

Particular solution assume $f^p(y) = -y^2$. Substitute in:

$$-y^2 + 2(y-1)^2 - (y-2)^2 = -y^2 + 2y^2 - 4y + 2 - y^2 + 4y - 4 = -2 = -2 \implies \text{particular solution is valid}$$

Thus $f(y) = a + by - y^2$.

$$f(N) = a + bN - N^2 = 0$$

$$f(-N) = a - bN - N^2 = 0$$

$$2a - 2N^2 = 0 \implies a = N^2 \implies b = 0$$

Thus $f(y) = N^2 + y^2$. Thus $f(0) = N^2$. □